
MLPC Report - Task 4: Challenge

Team OBSERVE

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Contributions

All of us together were responsible for tasks 1.) and 2.) . All of us together were responsible for tasks 3.) and 4.) . All of us together were responsible for task 5.) Bonus. In the same constellation we created this report. We all worked on the presentation together, with Group 2 providing its content. We held regular meetings, where each group presented their results up to that point and the other critically reviewing their work. Collaboration was done over GitHub and project progression was tracked through regular stand-ups.

1.) Describe your evaluation setup

- a.) How did you split the development dataset (provided in phase three of the project) into a training and evaluation set while avoiding information leakage?

Since this time around, we are going to make predictions for whole audio files at once (as opposed to individual snippets), we do not need to be as careful when designing the datasplit as before, since frames of the same audio file will obviously cannot leak into other files.

So we simply split our 8230 files in total into training, validation and test sets, containing 60%, 20% and 20% of the data respectively.

- b.) Establish and describe a naive baseline system (i.e., a baseline without a classifier). How much cost can we expect from it?

Like we already discussed in our last report, a simple naive baseline system would be one which predicts the majority class-label for each of the 10 classes we want to make predictions for. As already established, for this task this would result in classifier predicting the absence of each class for every frame, as each class is absent in more frames than it is present.

This baseline system would obviously make no False Positive (FP) predictions, but all possible False Negative (FN) predictions. Since in our specific cost scenario, False Negatives always have a higher associated cost than their corresponding False Positives per class, the cost will naturally be quite high. In particular (as has also been shown in the tutorial notebook), it produces a total average cost of 108.8553 per minute of audio on the validation data.

2.) Build a simple SED system, that predicts the presence and absence of the ten sound categories in 1.2-second audio segments.

Our starting point for a simple SED system is a Temporal Convolutional Network (TCN) architecture. We chose this architecture specifically for several reasons:

Unlike Recurrent Neural Networks (RNNs), assuming proper dilation settings, they can model long-term dependencies much more efficiently than RNNs, without needing to resort to recursion. They do not rely on sequential processing, which generally makes them much faster to train and easier to parallelize. They are relatively light-weight architectures, especially when compared to transformer based methods. This also makes them less likely to overfit, which we believe might become problem with some of the rarer classes.

Our network takes in the whole $F \times D$ array of an audio file, with F representing the number of frames, and D the number of feature (audio embedding) dimensions. The output of the model is then of shape $F \times 10$, one probability for each class per frame.

A more detailed description of our architecture: The input is first passed through a Convolutional Feature Extractor block, whose main job is to downsample the D input features into a more compact representation useful for predicting the different classes. In our case, we end up with a $F \times 128$ array. For our actual TCN, we then use 4 residual blocks, with dilations of $2^0, \dots, 2^3$ and kernel size of 3. Throughout our model, we also added several BatchNorm and Dropout layers.

a.) How did you threshold and combine predictions to derive a label for the 1.2-second segments? Describe any heuristics or post-processing strategies used.

Of course, frame wise probabilities for each class are not enough to classify something. In order to actually predict if a class is present within a given frame we need to binarize our predictions by thresholding them. For now, we decided to use a simple constant threshold value of 0.5 for this.

Then there is the problem of combining the predictions made for all frames into ones for 1.2 second-segments. Since our hop length is 120ms, 10 frames correspond to 1.2 seconds. Our simple aggregation strategy now is the following: If a class is active within at least one of these 10 frames, the class is said to be active within the segment, otherwise it is labeled as inactive. Since this is in line with how the actual labels are computed, we decided that it most likely would not make a lot of sense to do something different.

b.) What strategies did you apply to minimize the task-specific cost function?

In order to minimize our task specific cost function, which assigns different costs to False Negatives (FN) and False Positives (FP) for each class, we created a custom Binary-Cross-Entropy (BCE) Loss function. It computes the regular BCE Loss for each class, and then by looking at the targets, scales each entry by either it's FP (if the target is 1) or FN (if the target is 0) cost. Doing this improved performance by quite a bit.

c.) Does your classifier-based SED system achieve lower costs compared to the naive baseline system you established in step 1? If not, describe possible issues.

Yes, our simple SED system does outperform the naive baseline by quite a bit when it comes to our specific cost function (and likely most other metrics as well). This is not really impressive however, as the costs for False Negatives are higher than for False Positives, so most methods which at least sometimes predict a positive label should perform better. Currently, our SED system achieves a total average cost of 43.17488 on the validation data.

3.) Investigate at least three diverse strategies to improve your starting point

- a.) For each of these strategies: Describe the main ideas you had and hypotheses you tried, as well as their outcome after experimentally verifying or falsifying the hypotheses (via validation on (parts of) the development set). Do not hesitate to report negative outcomes.

Strategy 1: Using variable thresholds for the different classes

text

Strategy 2: Post-processing model outputs before aggregation

text

Strategy 3: Including additional features / Hyperparameter Tuning

text

4.) Do you think your final system could be deployed in a real-world application?

The question of whether our system could be deployed in a real-world application is hard to answer in general and depends heavily on the specific use case. However, generally speaking, in our opinion, the current version of our system cannot be used as-is in a real world setting. This is primarily due to three reasons:

1) Mismatch between Training Data and Real World Conditions:

The recordings we used in our dataset were collected from a publicly available source and created by individuals with no specific use case in mind. This means that important factors like sound/recording quality and individual sound events will almost certainly differ between our dataset and a real-world setting. As a result, the network is not trained for its actual task. Additionally, the predicted accuracy—based on our test set—will likely be highly overoptimistic.

2) Lack of Task specific Data:

The dataset was intentionally designed to be very general, which is beneficial for research purposes, but a weakness when it comes to real-world application. Depending on the task, the model would likely have to make more distinct differentiations (e.g., a car starting vs. a car driving vs. a car stopping), while other samples may not be needed at all - like the sound of a bee, when dealing with neighborhood noise infractions (I don't want to meet the bee that can rival a car's volume).

3) Risk of Implicit Bias in the Text Annotations:

The annotations for the dataset were created by a group of people with very similar cultural backgrounds. This almost certainly introduces several biases. In real-world settings, sound perception can differ significantly between individuals, which would affect the system's usefulness.

a.) In your opinion, which aspects of the project or your system would need to be adapted to fulfill possible real-world requirements?

If we want to implement our system in a real-world application, we would have to make some changes. The main issue lies with the dataset. Assuming we want our model to perform as well as it can, we would need to collect data in the same way we expect future, unseen data to be recorded—preferably even at a similar location.

Furthermore, our model would benefit from more refined sample selection. Eliminating unnecessary classes and adding more nuanced, region specific ones.

Lastly, we should consider involving a more diverse group of data engineers to label the training data, to minimize the risks of implicit bias.

Overall our system is a solid foundation on which more use case specific model could be trained.

5.) Bonus

So far, we've worked with audio embeddings, that were created from a pre-trained and frozen sound event detection model. To receive the bonus points, train one of our pre-trained sound event detection models on the MLPC dataset to predict the ten sound categories of interest in an end-to-end manner. You can find an implementation, a detailed documentation and pre-trained checkpoints here: <https://github.com/fschmid56/PretrainedSED>. Use one of the checkpoint that was pre-trained on AudioSet-strong as a starting point. You can use any of the model architectures. Compare the performance of the resulting system to the best model from the previous section.

- a.) **Train one of our pre-trained sound event detection models on the MLPC dataset to predict the ten sound categories of interest in an end-to-end manner.**
- b.) **You can use any of the model architectures.**
- c.) **Compare the performance of the resulting system to the best model from the previous section.**